

Test 3 — Question Paper

38 items. Suggested time: 95 minutes (about 2 minutes 30 seconds per item).

A four-function calculator may be used **only** on items marked *calculator*.

The Grade 6 B.E.S.T. Mathematics Reference Sheet is on the last page.

Aligned to the FAST Grade 6 Mathematics test design: 33–42% Number Sense and Operations, 25–36% Algebraic Reasoning, 25–36% Geometric Reasoning, Data Analysis, and Probability.

1. A data set is shown.

0, 2, 2, 3, 4, 4, 4, 6, 6, 8, 11, 17

A histogram is made with the intervals 0-4, 5-9, 10-14 and 15-19.

How tall is the bar for the interval 10-14?

- A. 11
- B. 2
- C. 1
- D. 3

Show your work

multiple choice

2. Sofia had some stickers and gave away 19 of them. Now Sofia has 34 stickers.

Which equation can be used to find x , the number of stickers Sofia started with?

- A. $x - 19 = 34$
- B. $19 - x = 34$
- C. $x + 19 = 34$
- D. $x + 34 = 19$

Show your work

multiple choice

3. A bus travels 240 miles in 4 hours at a constant speed.

At this rate, how far does the bus travel in 1 hour?

- A. 960 miles
- B. $\frac{1}{60}$ miles
- C. 236 miles
- D. 60 miles

Show your work

multiple choice

4. A box shaped like a right rectangular prism has a volume of 96 cubic centimeters. Its base has an area of 32 square centimeters.

What is the height of the box, in centimeters?

- A. 3
- B. 6
- C. 32
- D. 64

Show your work

multiple choice

5. Ximena recorded these scores:

3, 4, 4, 5, 6, 10, 18

What is the median of the data set?

- A. 5
- B. 7
- C. 15
- D. 4

Show your work

multiple choice

6. What is the value of $8(11 + 7)$?

- A. 616
- B. 144
- C. 26
- D. 95

Show your work

calculator · multiple choice

7. What is $1.25 \div 0.25$?

- A. 0.5
- B. 50
- C. 0.3125
- D. 5

Show your work

calculator · multiple choice

8. What value of x makes the equation $\frac{2}{3} \times x = \frac{2}{15}$ true?

- A. $\frac{4}{5}$
- B. $\frac{1}{5}$
- C. $\frac{4}{45}$
- D. $\frac{10}{3}$

Show your work

multiple choice

9. Which expression is equivalent to $2x + 12 + 4x$?

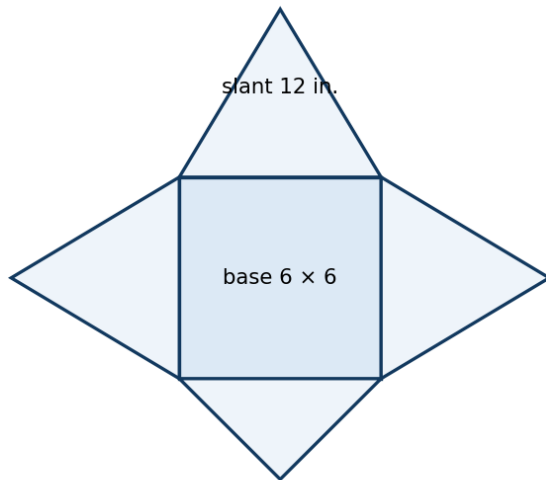
- A. $8x + 12$
- B. $6x + 12$
- C. $6x + 12x$
- D. $18x$

Show your work

multiple choice

10. The net of a right rectangular pyramid with a square base is shown.

What is the surface area of the pyramid, in square inches?



- A. 36
- B. 180
- C. 288
- D. 324

Show your work

multiple choice

11. A data set is shown.

10, 11, 11, 14, 19

The value 26 is added to the data set. What is the range of the new data set?

- A. 9
- B. 19
- C. 26
- D. 16

Show your work

multiple choice

12. What is the value of $-29 - (-22)$?

- A. -4
- B. -51
- C. 7
- D. -7

Show your work

calculator · multiple choice

13. Which value of x makes the inequality $x > 19$ true?

- A. 13
- B. 17
- C. 19
- D. 22

Show your work

multiple choice

14. Buses on one route leave Government Center every 36 minutes and buses on another route leave every 60 minutes. Both leave at 9:00 a.m.

In how many minutes will they next leave at the same time?

- A. 12
- B. 2160
- C. 180
- D. 96

Show your work

calculator · multiple choice

15. Store A sells 8 bottles of water for \$4.00. Store B sells 16 bottles for \$9.00.

What is the unit price at Store A, in dollars per bottle?

- A. 0.5
- B. 12
- C. 0.5625
- D. 2

Show your work

multiple choice

16. The table shows equivalent part-to-part-to-whole ratios.

What number belongs in the box marked ?

Boys	Girls	Total
5	6	11
15	18	?

Part-to-part-to-whole ratios

- A. 21
- B. 36
- C. 33
- D. 11

Show your work

multiple choice

17. Use the drop-down menu to complete the sentence.

The expression $10x$ represents the product of 10 and a number x .

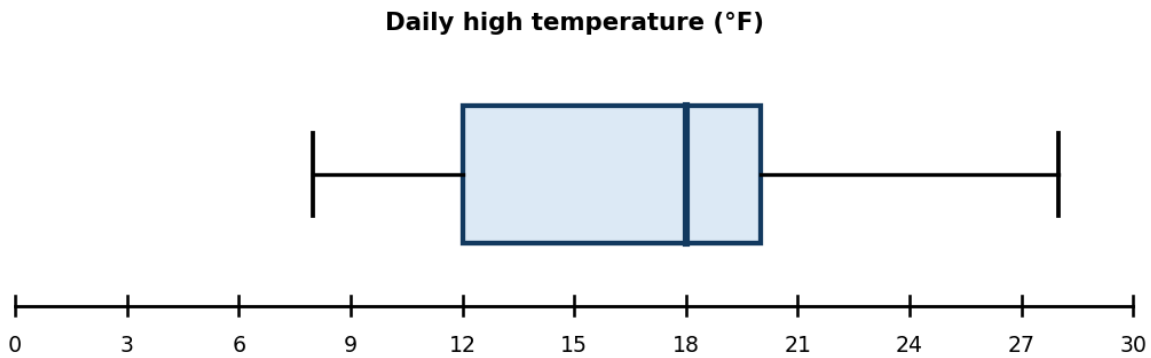
Blank (1) choices: $10x$ | $x + 10$ | $x - 10$

Show your work

editing task choice

18. The box plot shows a set of data.

What is the interquartile range of the data?



- A. 10
- B. 18
- C. 8
- D. 20

Show your work

multiple choice

19. What is the value of $-45 \div (9)$?

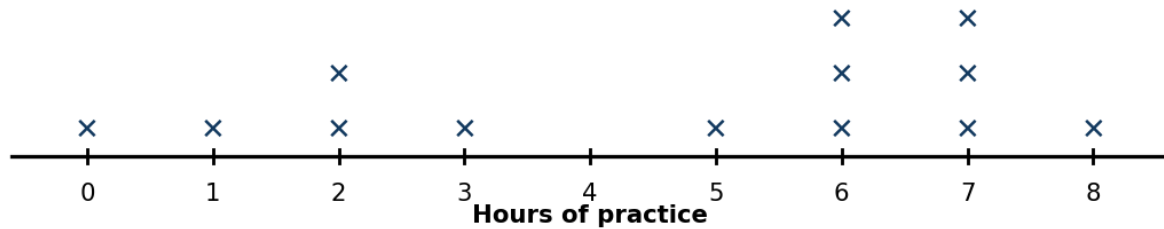
- A. 5
- B. -54
- C. -5
- D. 9

Show your work

calculator · multiple choice

20. The line plot shows hours of practice.

What is the range of the data?



- A. 10
- B. 9
- C. 13
- D. 8

Show your work

multiple choice

21. Andres pays with \$100 for 5 items that cost \$6.13 each.

How much change should Andres receive?

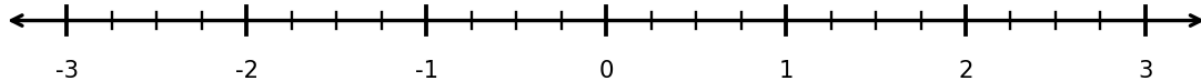
- A. \$93.87
- B. \$69.35
- C. \$75.48
- D. \$30.65

Show your work

calculator · multiple choice

22. Plot these three numbers on the number line.

-2, $-\frac{1}{4}$, $\frac{1}{2}$



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

23. Tomas shared 99 guavas equally among 11 baskets.

Which equation can be used to find x , the number of guavas in each basket?

- A. $x + 11 = 99$
- B. $x \div 11 = 99$
- C. $11x = 99$
- D. $99x = 11$

Show your work

multiple choice

24. What is $1\frac{3}{4} \div \frac{5}{7}$?

- A. $2\frac{9}{20}$
- B. $\frac{20}{49}$
- C. $1\frac{1}{4}$
- D. $2\frac{13}{28}$

Show your work

calculator · multiple choice

25. A bag holds 25 mangoes and 30 guavas.

What is the ratio of mangoes to the total number of pieces of fruit, in simplest form?

- A. 5:11
- B. 11:5
- C. 6:5
- D. 5:6

Show your work

multiple choice

26. Select **all** the values of x that make $x \leq 48$ true.

Circle every correct choice:

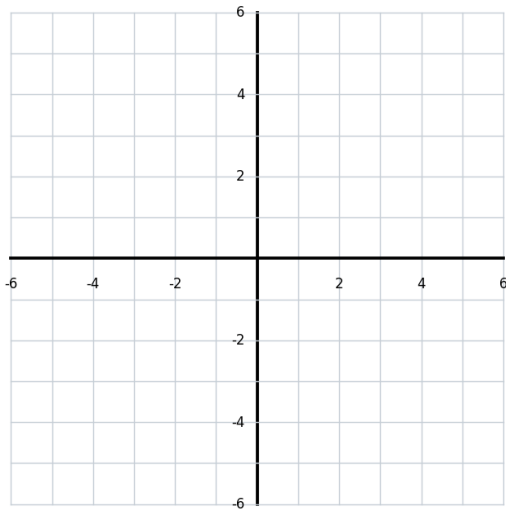
- ☐ 44
- ☐ 47
- ☐ 48
- ☐ 49
- ☐ 53
- ☐ 57

Show your work

selectable hot text

27. Plot these three points on the coordinate plane.

$(-5, -5)$, $(5, -5)$, $(-5, 5)$



Mark your answer on the diagram above.

Show your work

graphing (grid)

28. What is the value of $(\frac{2}{5})^2$?

- A. $\frac{4}{25}$
- B. $\frac{2}{5}$
- C. $\frac{2}{25}$
- D. $\frac{4}{5}$

Show your work

calculator · multiple choice

29. Which fraction is equivalent to 80%, written in simplest form?

- A. 8
- B. $\frac{8}{5}$
- C. $\frac{5}{4}$
- D. $\frac{4}{5}$

Show your work

calculator · multiple choice

30. A number has the prime factorization $2^2 \times 3^2 \times 11$.

What is the number?

- A. 396
- B. 792
- C. 21
- D. 198

Show your work

calculator · multiple choice

31. What is the value of $|-17| - |10|$?

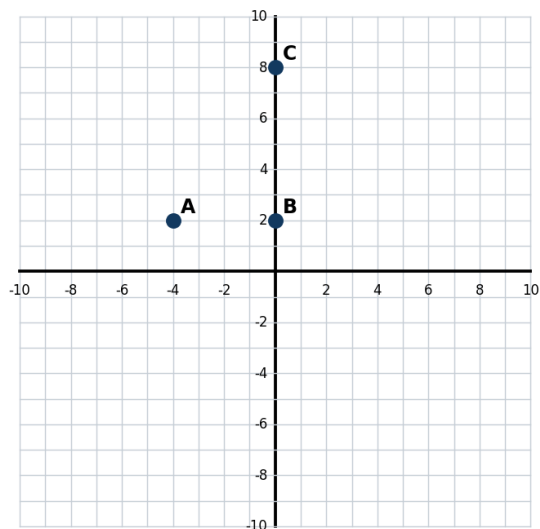
- A. 7
- B. 9
- C. -7
- D. 27

Show your work

calculator · multiple choice

32. Points A, B and C are three vertices of a rectangle.

Plot the fourth vertex on the coordinate plane.



Mark your answer on the diagram above.

Show your work

graphing (grid)

33. A point is graphed at $-4\frac{3}{4}$ on a number line.

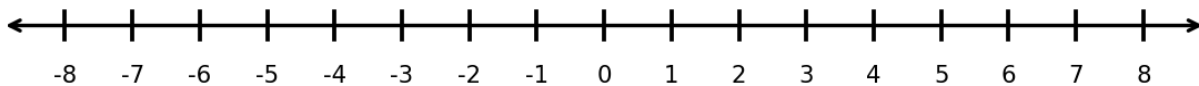
How many units is the point from 0?

- A. $-4\frac{3}{4}$
- B. $4\frac{3}{4}$
- C. $5\frac{3}{4}$
- D. $2\frac{3}{8}$

Show your work

calculator · multiple choice

34. A football team gained 4 yards on one play and lost 2 yards on the next. Plot the two integers that represent these two plays, and plot 0.



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

35. 16 students is 40% of the students in a grade.

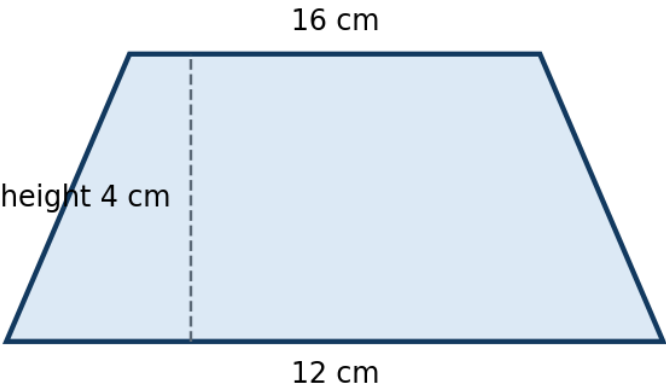
How many students are in the grade?

- A. 40
- B. 1600
- C. 640
- D. 56

Show your work

multiple choice

36. What is the area of the trapezoid shown, in square centimeters?

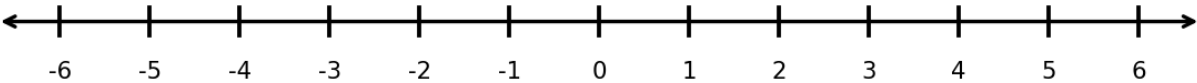


- A. 56
- B. 112
- C. 48
- D. 16

Show your work

multiple choice

37. Plot the two numbers on the number line that are exactly 3 units from 0.



Mark your answer on the diagram above.

Show your work

calculator · graphing (grid)

38. Which question is **not** a statistical question?

- A. How many books did each student in the class read last month?
- B. How old is my teacher?
- C. How many minutes did each sixth grader spend on the bus this morning?
- D. What are the heights, in inches, of the students in the band?

Show your work

multiple choice

Grade 6 Mathematics Reference Sheet

Customary Conversions

1 foot = 12 inches	1 cup = 8 fluid ounces
1 yard = 3 feet	1 pint = 2 cups
1 mile = 5,280 feet	1 quart = 2 pints
1 mile = 1,760 yards	1 gallon = 4 quarts
1 pound = 16 ounces	1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters	1 liter = 1,000 milliliters
1 meter = 1,000 millimeters	1 gram = 1,000 milligrams
1 kilometer = 1,000 meters	1 kilogram = 1,000 grams

Time Conversions

1 minute = 60 seconds	1 week = 7 days
1 hour = 60 minutes	1 year = 365 days
1 day = 24 hours	1 year = 52 weeks

Formula

Rectangular prism	$V = lwh$ or $V = Bh$
l = length, w = width, h = height	B = area of base, V = volume